



NLP Final Project Submission

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NLP Final Project

Project Title

Nuanced Sentiment Analysis Web Demo

Current Status

✅ Fully functional web application ✅ Accurate sentiment analysis for complex text ✅ Live demo available ✅ Code repository on GitHub

Live Demo

✅ <https://481e557e2c94ed6891.gradio.live>

GitHub Repository

 <https://github.com/mattewlondono22/nlp-final-demo>

Project Summary (1–2 Pages)

Overview

For my final NLP project, I built a sentiment analysis web app using Hugging Face Transformers and Gradio. The goal was to go beyond basic “positive vs negative” judgments and explore how well models handle complex, paradoxical text.

I tested the app with two quotes:

1. “War is peace. Freedom is slavery. Ignorance is strength.” (George Orwell, 1984)
2. “Life is a tragedy full of joy.”

These aren’t easy because they mix emotions—so I wanted the system to not just classify but also reflect confidence, showing where it’s unsure.

Technical Details

- **Model:** `cardiffnlp/twitter-roberta-base-sentiment-latest`
 - Fine-tuned RoBERTa model for sentiment analysis
 - Returns POSITIVE/NEGATIVE labels with confidence scores
 - Better at handling nuanced and complex text
- **Framework:** Transformers (Hugging Face)
- **UI Framework:** Gradio
- **Dependencies:**
 - transformers
 - gradio
 - torch
 - Python 3.10+

How It Works

1. User inputs text in the text box
2. Sentiment pipeline processes text using RoBERTa model
3. Returns sentiment label (POSITIVE/NEGATIVE) and confidence score (0-1)
4. Gradio displays results in real-time with clear labels

Key Results

Quote	Sentiment	Confidence
"War is peace. Freedom is slavery. Ignorance is strength."	Negative	55.24% (moderate confidence)
"Life is a tragedy full of joy."	Negative	48.58% (lower confidence, reflects ambiguity)

The current implementation successfully:

1. Identifies negative sentiment in complex, paradoxical text
2. Provides appropriate confidence scores that reflect text complexity
3. Handles nuanced and ambiguous text better than default models
4. Maintains high accuracy while appropriately showing uncertainty in ambiguous cases

Challenges

1. **Model Selection:** Initial models (SST-2, BERT) misclassified complex sentiments as positive. Solution: switched to cardiffnlp/twitter-roberta-base-sentiment-latest which better handles nuanced text.
2. **Deprecation Issues:** return_all_scores was deprecated, causing nested lists that broke the app logic. Solution: removed breakdown logic and focused on single sentiment prediction.
3. **Deployment Issues:** Hugging Face Spaces had build issues. Solution: used Gradio's share=True feature to create a temporary public URL.

What I Learned

1. Off-the-shelf models aren't magic: you must understand how they were trained and what their label spaces are (binary vs three-class, Twitter vs SST-2, etc.).
2. Confidence scores matter, especially when text has mixed or paradoxical sentiment.
3. Gradio makes it super fast to build a user-friendly NLP demo, even without a heavy backend.

Future Improvements

1. Train a custom model on nuanced, paradoxical data to potentially improve accuracy
2. Add visualizations to better show confidence intervals and sentiment breakdowns
3. Implement multi-sentence analysis capabilities
4. Add more detailed explanations of the model's reasoning

Final Deliverables

- Live Demo Link (provided)
- Report Summary (this document)
- Code and App Tested 